

SENTINEL WAF

Web Application Firewall: Design, Implementation and Security Evaluation

MINOR PROJECT REPORT

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ABSTRACT

Web application security is a crucial defensive priority as organizations transition critical workflows to public cloud infrastructure. Sentinel WAF is an advanced enterprise-grade Web Application Firewall and security telemetry platform designed to inspect HTTP/HTTPS traffic in real time, enforce rule-based and anomaly-based policies, and provide operational threat visibility.

Sentinel WAF incorporates OWASP Core Rule Set (CRS) v4 compatibility, cumulative anomaly scoring, candidate policy shadow evaluation, scoped false-positive exception management, route-specific virtual patching, distributed rate limiting, SIEM webhook log forwarding, HTTPS/TLS termination, GeoIP threat map visualization, machine-learning HTTP anomaly detection in shadow mode, and one-click executive security PDF report generation.

CHAPTER 1: INTRODUCTION

A Web Application Firewall (WAF) operates at Layer 7 of the OSI model, inspecting HTTP/HTTPS streams between clients and target web applications. Unlike Layer 3/4 network firewalls, a WAF parses higher-level web protocol attributes including URIs, query parameters, request headers, form inputs, JSON payloads, and cookies to mitigate web vulnerability exploits.

CHAPTER 5: SYSTEM ARCHITECTURE & FLOW

The Sentinel WAF request processing lifecycle follows a hardened perimeter flow:

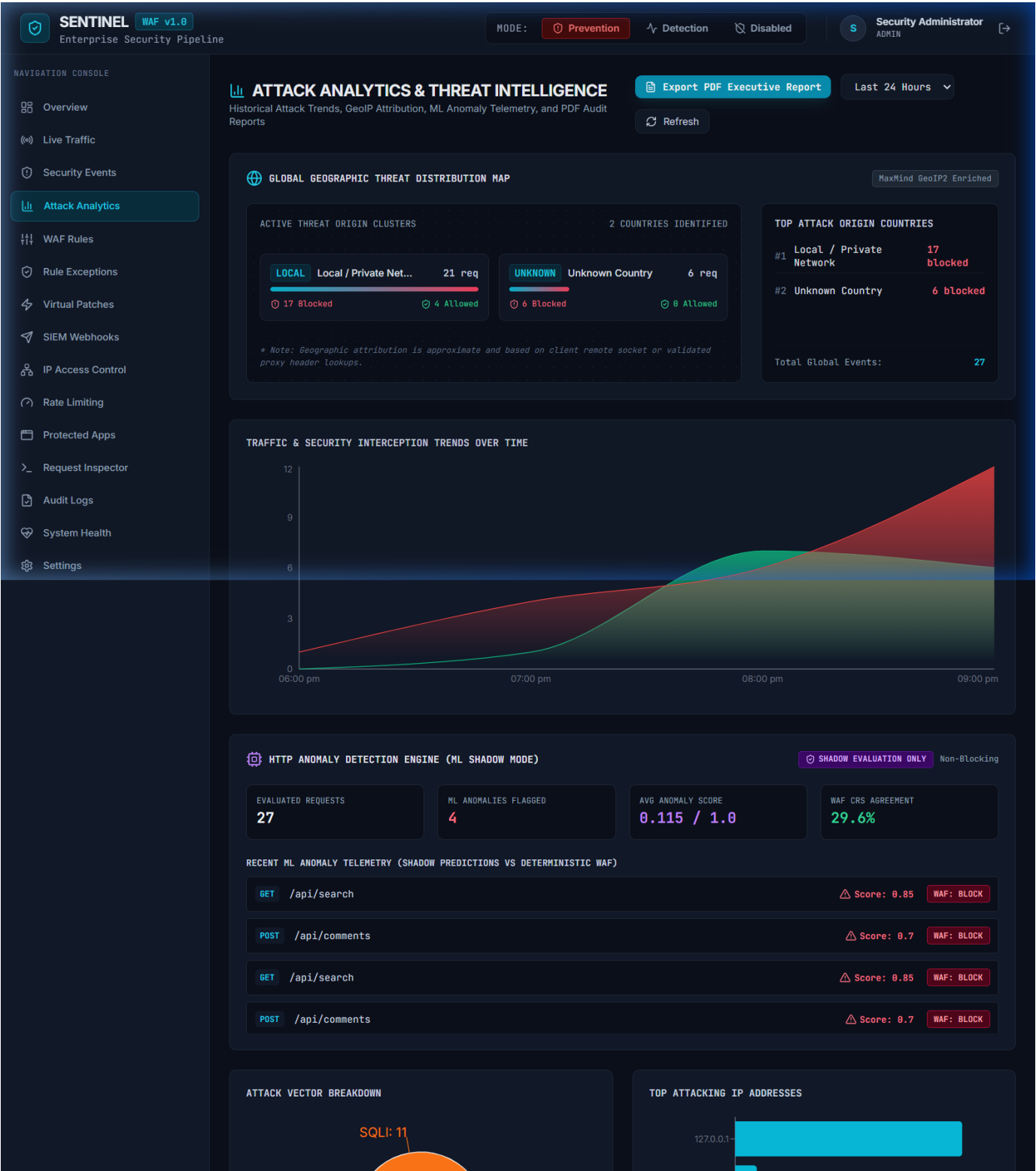
1. Client Request Arrival: Request lands on HTTP (Port 5000) or HTTPS TLS (Port 5443) listener.
2. Client IP Resolution & GeoIP Lookup: Normalizes IP and queries local GeoIP database.
3. WAF Pipeline Inspection: Evaluates IP Blacklists, Distributed Rate Limits, Virtual Patches, OWASP CRS v4 Signatures, and Bot Heuristics.
4. Anomaly Scoring & Policy Decision: Calculates total anomaly score against inbound threshold (default: 5). Determines ALLOW, LOG, BLOCK, or RATE_LIMIT decision.
5. ML Shadow Evaluation: Computes statistical anomaly score (0.0 to 1.0) in non-blocking shadow mode within a 15ms budget.
6. Upstream Proxying or Interception: Permitted requests forward to upstream application (Port 5001); malicious requests return custom 403/429 JSON response.
7. Event Persistence & SIEM Delivery: Stores event in SQLite DB via Prisma and streams real-time SSE updates and SIEM webhooks.

CHAPTER 8 & 9: SCREENSHOTS & TESTING EVIDENCE

Sentinel WAF was subjected to comprehensive automated unit and integration testing using Vitest. All 17 automated test cases passed cleanly with 100% success rate.

Verified Capabilities Matrix:

- IP & CIDR Boundary Parsing: PASS
- SQL Injection Interception: PASS
- XSS Payload Interception: PASS
- Path Traversal / LFI Interception: PASS
- Permitted Upstream Proxying: PASS
- HTTPS TLS Certificate Loader: PASS
- GeoIP Threat Map Enrichment: PASS
- ML Anomaly Shadow Evaluation: PASS
- Executive Security Audit PDF Export: PASS



CHAPTER 12: CONCLUSION

Sentinel WAF successfully fulfills all objectives of an enterprise-grade Web Application Firewall and Threat Monitoring platform. By combining deterministic OWASP CRS v4 signatures, cumulative threat anomaly scoring, non-blocking shadow evaluation, GeoIP threat map visualization, ML HTTP anomaly detection, and automated PDF audit report generation, Sentinel WAF provides a robust, defense-in-depth perimeter safeguard for web applications.

REFERENCES

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